Project Plan

Dormio

Industry Partner	
Primary Instructor	Laily Ajellu
Team Member	Bryan Paul Yabut
Team Member	Ibrahim Abdillahi
Team Member	Patrick Sherwin Millares
Team Member	Patrick Solamillo
Team Member	Yousuf Salihi

Document Revision History

Revision #	Date
2.5	**9/29/25**
3.0	**1/23/26**
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1. Executive Summary

Objective	To design and develop an android app called Dormio. Dormio is built with the objective of simplifying daily resident life for students by automating and centralizing their tasks, and to provide residence staff with an efficient system for maintenance management.
Corporate Goals Addressed	• Enhance student well-being and success - by centralizing management of schedules, budgets and chores, students can better focus on their academics. • Improve residence management efficiency • Modernize the campus technology ecosystem • Optimize maintenance operations and reduce administrative overhead of Residence Staff
Planned Start Date	January 12, 2026
Planned End Date	March 16, 2026

2. Project Approvers, Reviews and Distribution List

Approvers, reviewers and distribution list

Project Role	Name	E-mail	Date
Project Manager	Laily Ajellu	Laily.Ajellu@georgebrown.ca	1/23/26
Back-end Developer	Bryan Paul Yabut	bryn.ybt@gmail.com	1/23/26
Front-end Developer	Ibrahim Abdillahi	ibrahimabdillahi2222@gmail.com	1/23/26
Back-end Developer	Patrick Sherwin Millares	millareslepatrick@gmail.com	1/23/26
Back-End Developer	Patrick Solamillo	Patricksolamillo@gmail.com	1/23/26
Front-End Developer	Yousuf Salihi	salihiyousuf@gmail.com	1/23/26

3. Scope

Define the sum total of all of its products and their requirements or features.

In Scope	Out of Scope *
User and Staff Documentation - Provision of user guides and technical documentation that clearly illustrate application flow, features, and system architecture	**Live Payment Processing** - Dormio will not process, integrate with, or initiate any online transactions.
User Management - Implementation of secure user accounts, profiles and role based access for both student and residence staff	**Handling of Sensitive Information** - Dormio will not store sensitive data such as financial credentials.
Customizable Meal Manager - functionality allowing students to select, save, and modify a personalized, meal plan based on preferences.	**Migration of Data** - there will be no automated tools to migrate existing student information into Dormio

Budget Management System - A tool for personal finance tracking, including manual transaction entry and bill splitting functionality among assigned roommates.	**Third-Party Software Integration** - Dormio will not integrate third-party software.
Maintenance Requests System - Allows students to submit and track maintenance requests, and enables Residence staff to manage and update request status.	**Group Communication Features** - features such as in-app group chat or direct messaging between residents are out of scope
Chore Assignment - Allow users to create, track and assign chores with a section allocated to quickly review any upcoming tasks	**External Calendar Synchronization -** Dormio will not include Syncing the calendar with external systems like google calendar and outlook
User Notification - User will be sent firebase-based alerts to notify them of any updates or changes	**Kitchen Inventory Management -** Dormio will not have the ability to track ingredient stock levels or handle supplier management
Calendar System - User will have access to a calendar where user can view any upcoming classes/work or other events in a single place	**Local Data Viewing -** The application is completely reliant on connection to the server and no data is stored locally to view if servers are down

4. Deliverables

This project will deliver the following.

Deliverable	Description
Project Plan	This Documentation will outline how our team will execute the creation and management of the Dormio Project. It includes management of members' tasks, timeliness and risk management strategies as well as providing a road map to track progress and meet deadlines.
Team Charter	Team charter defines roles and our purpose/responsibility with what needs to be done by each member. It actively helps with ensuring collaboration and communication keeping all parties accounted for their work through development.
Product Backlog	A comprehensive, prioritized list of all desired features, requirements, and enhancements for the Dormio application
Sprint Backlog	A subset of items selected from the Product Backlog, along with the team's plan for implementing them during a specific, upcoming sprint cycle.
Gantt log Chart	A visual timeline that illustrates the project schedule, showing the planned start and end dates for all tasks (WBS) and milestones, clearly displaying dependencies.
Demonstration and Final Presentation	Demonstration will show the usage of our project with detailed explanations of its design and functions.
Backend API Service	The Node.js/express server contains most of the business logic
Full feature Dormio app	The Dormio app will be created and fully functioning with features created as described in the scope
User and Staff Documentation	User and Staff Documentation will be provided and accessible inside of the Dormio app. It will contain information

	on how to use the Dormio app in detail.

5. Assumptions

This project makes the following assumptions;

1. **Consistent User Connectivity**: We assume student residents will have reliable and consistent internet access within the dormitories, as the app app requires continuous connectivity for real time updates(FCM) and database synchronization

2. **Administrative System Adoption**: We assume the Dorm Administration team will fully adopt and integrate the new Maintenance Request System, and provide any necessary access permissions for the system to interface with residence workflows.

3. **Data Security Sufficiency**: We assume the security measures implemented are fully sufficient and compliant to adequately protect all sensitive student information, particularly financial and personal data.

4. **Positive User Engagement**: We assume the target student user group will be willing to consistently and frequently use the new app for managing daily tasks like finances and meal planning.

5. **Technical Stack Stability**: We assume the chosen technology stack will remain accessible (free or within budget) and stable throughout the development and testing lifecycle.

6. **Unchanged Project Scope:** We assume the finalized set of features and requirements will remain fixed after the initial approval, preventing any scope creep that would jeopardize the project deadline.

7. **Resource Availability**: We assume all core team members will maintain the agreed-upon availability and commitment needed to complete their assigned tasks without major unforeseen personal or academic conflicts.

8. **Environment details:** JWT secret keys, database URLs, and FCM credentials are configured

6. Dependencies

The following are the internal and external dependencies that will have to be acknowledged and addressed;

1. **Database Connection & Seeding:** Our software is dependent on the population of the database. All feature functionality and testing are blocked until the database is successfully connected and populated with initial test data.

2. **Version Control System (VCS) Setup:** Our team will push to the shared Git repository so that each member is working on the latest version of the software. All collaborative coding efforts are dependent on this system being properly set up.

3. **Finalized Design Documents:** Successful implementation of features is dependent on the final sign-off of the Wireframes for all features and the Designing database schemas and relationships.

4. **Team Member Output:** Implementation of complex features is dependent on the successful completion of foundational work by other team members

5. **Administrative Access Credentials:** Final testing and deployment of the Maintenance Request System requires cooperation and the provision of necessary access credentials from the Residence Staff.

6. **Required External Approvals:** The implementation of features impacting staff (like the Maintenance Request System) is dependent on final approval and sign-off from the Residence Staff/Management on the proposed workflow and data collection.

7. **Resource Allocation:** The project timeline is dependent on all team members maintaining their commitment to the hours allocated in the RAM and Gantt chart.

8. **Firebase Cloud Messaging:** FCM is required for the push notification functionality for Dormio

7. Risk Management

Potential Risk	Severity (H/M/L)	Likelihood (H/M/L)	Management Strategy
Team member leaves the group	M	M	Tasks are broken up into multiple smaller tasks. These tasks are then divided up amongst remaining members. Potentially request new members.
Team member is unable to complete work by deadlines	L	L	Tasks are divided amongst other members. Participation is marked and if the issue continues it can be escalated to the professor.
Scope of the project is greater than previously thought.	H	M	Create a buffer of time for task completion (e.g 5 days before deadline) Simplify features to bare minimum required. Additional research of code libraries that may aid in completion. Project plan and timeline need to be updated.

Third party service such as database becomes unavailable/paid	M	L	If the database is down, prevent usage of features that require database functionality.
Loss of projects codes and files	H	L	Create backups regularly and create plans for team members to follow for continuous usage of the version control system.
Features work on one person's computer but not on others	M	M	Ensure that dependencies are added to version control to ensure the same dependencies are used. Environments such as emulators are created to be as consistent as possible according to a created project guide.
Features as described in Project Reports are not detailed enough causing unclear requirements for feature	L	L	During meetings when working on code. The full extent of the features will be discussed. What features are the focus of a

			meeting will be decided by the Gantt chart
Team member lacks technical knowledge required	M	M	Tasks can be divided depending on technical knowledge. Since tasks are divided by project role their partner can assist them if necessary.
Code/ Features from separate members do not work once integrated together	H	M	The structure and necessary objects will be defined early in the project. Builds will be frequently integrated(Weekl y) so that any issues can be dealt with earlier on in the process.

8. Communication

Reporting

The following reports will be produced;

Report	Audience	Frequency
Project Vision	Project Manager	One-Time – (Sprint 1)
Business Requirements	Project Manager	One-Time – (Sprint 1)
Personas	Project Manager	One-Time – (Sprint 1)
User Stories	Project Manager	One-Time – (Sprint 1)
Minutes of Meeting	Project Manager	Weekly – Tuesday at 3:00:pm
Project Status Report	Project Manager	One-Time - (Sprint 5)

Meetings

The following meetings/communication will be established;

Meeting	Purpose	Attendees	Frequency
Meeting #1	Division of Work and approval of system features	Bryan Paul Yabut, Yousuf Salihi, Patrick Sherwin Millares, Patrick Solamillo, Ibrahim Abdaillahi	Weekly – Thursday at 1:30
Meeting #2	Creation of current sprint files. Tracking pace of current work.	Bryan Paul Yabut, Yousuf Salihi, Patrick Sherwin Millares, Patrick Solamillo, Ibrahim Abdaillahi	Weekly – Thursday at 1:30
Meeting #3	Division of Work for sprint 2	Bryan Paul Yabut, Yousuf Salihi, Patrick Sherwin Millares, Patrick Solamillo, Ibrahim Abdaillahi	Weekly – Thursday at 1:30

Meeting #4	Tracking current progress on sprint #2 and division of presentation responsibilities	Bryan Paul Yabut, Yousuf Salihi, Patrick Sherwin Millares, Patrick Solamillo, Ibrahim Abdaillahi	Weekly – Thursday at 1:30
Meeting #5	Updating trello board with requirements and foundational phase 1 setup (foundational phase)	Bryan Paul Yabut, Yousuf Salihi, Patrick Sherwin Millares, Patrick Solamillo, Ibrahim Abdaillahi	Weekly - Tuesday at 3:00

9. Task Listing (WBS- Work Breakdown Structure)

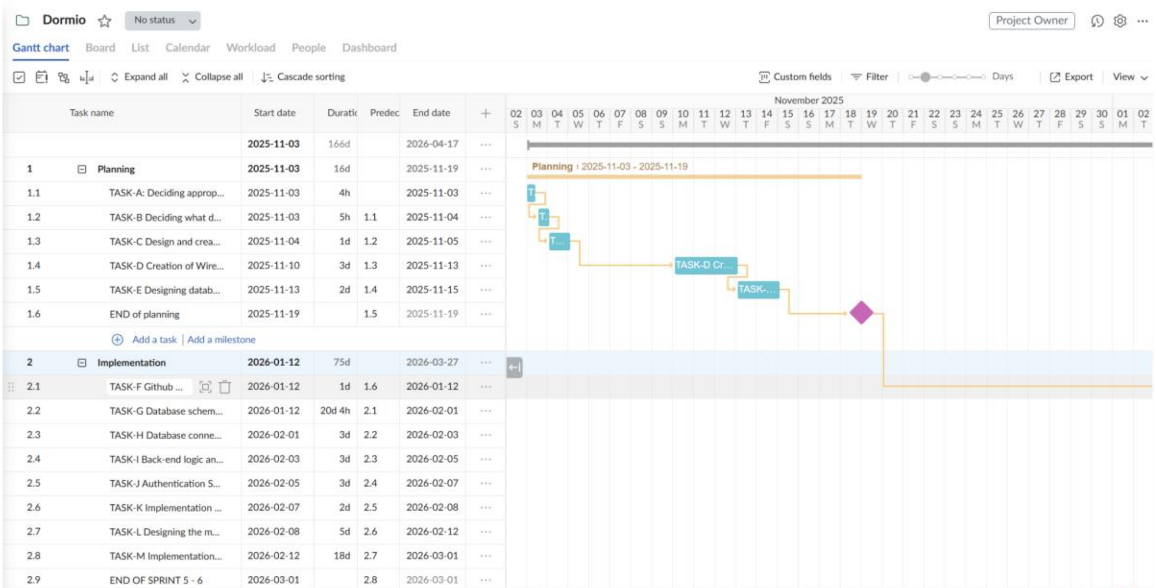
The following resource proposal template summarizes the resource hours committed to this project, upon final approval of this document.

Reference	Tasks	Duration	Dependency
A	Deciding on an appropriate tech stack for the application	4 hours	
B	Deciding what data needs to be stored locally and, on the database,	5 hours	A
C	Designing and creation of user roles and permissions in the system	8 hours	B
D	Creation of wireframes for all features (Mobile app)	3 days	C
E	Designing database schemas and relationships between tables in a SQL RDBMS	2 days	D
F	Setting up GitHub repositories and initializing the project for both server and Mobile app	4 hours	
G	Database schema rework using prisma and PostgreSQL	3 weeks	E
H	Setting up database connection and seeding initial data	1 hour	G
I	Backend Logic and Validation	2 days	F, G, H
J	Implementation of Authentication System	2 days	I
K	Implementation of CRUD operations for all features based on user roles	1 week	J
L	Designing the mobile app UI in simplified form	2 weeks	F, K

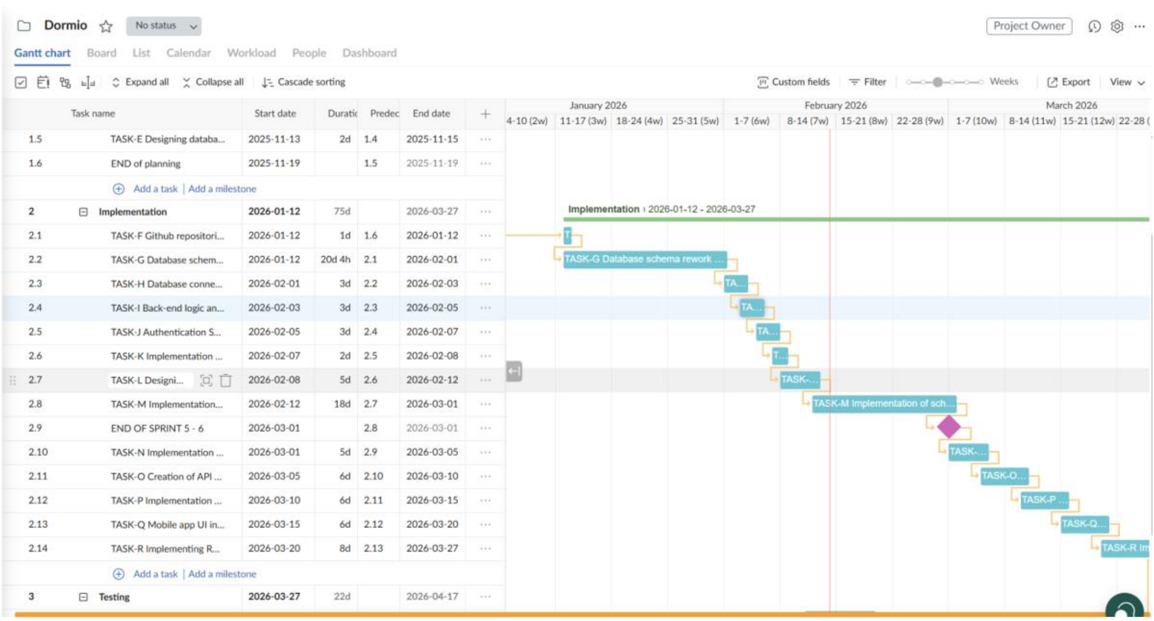
M	Implementation of scheduling, meal planning, and maintenance request features with logic to restrict choices	2 weeks	J
N	Implementation of Chore Manager and CSV export logic	3 days	M
O	Creation of API documentation for mobile app team	5 days	M, N
P	Implementation of data visualization for the Budget Management System	5 days	N
Q	Mobile app UI in complete form	5 days	P
R	Implementing REST API connection and integrate as features are built	1 week	K, L, M, N, O, P, Q
S	Integrate and test all implemented features into a unified dashboard	1 week	R
T	Final testing and debugging	1 week	S
U	Preparation for presentation and demo	1 week	T

10. Gantt Chart

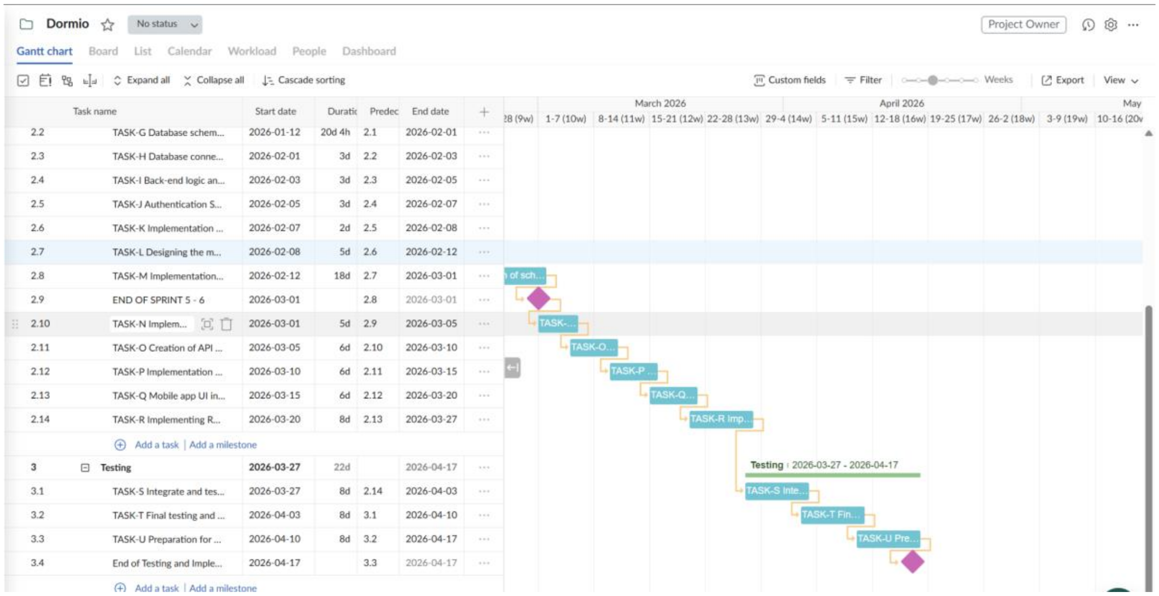
Link: Dormio Gantt chart PDF



SPRINT 5 - 6



SPRINT 7 - 8


Dormio
No status
Project Owner
Gantt chart
Board
List
Calendar
Workload
People
Dashboard
Expand all
Collapse all
Cascade sorting
Custom fields
Filter
Weeks
Export
View
Task name
Start date
Duratic
Predec
End date
March 2026
April 2026
May
2.2 TASK-G Database schem... 2026-01-12 20d 4h 2.1 2026-02-01
2.3 TASK-H Database conne... 2026-02-01 3d 2.2 2026-02-03
2.4 TASK-I Back-end logic an... 2026-02-03 3d 2.3 2026-02-05
2.5 TASK-J Authentication S... 2026-02-05 3d 2.4 2026-02-07
2.6 TASK-K Implementation ... 2026-02-07 2d 2.5 2026-02-08
2.7 TASK-L Designing the m... 2026-02-08 5d 2.6 2026-02-12
2.8 TASK-M Implementation... 2026-02-12 18d 2.7 2026-03-01
2.9 END OF SPRINT 5 - 6 2026-03-01 2.8 2026-03-01
2.10 TASK-N Implem... 2026-03-01 5d 2.9 2026-03-05
2.11 TASK-O Creation of API ... 2026-03-05 6d 2.10 2026-03-10
2.12 TASK-P Implementation ... 2026-03-10 6d 2.11 2026-03-15
2.13 TASK-Q Mobile app UI in... 2026-03-15 6d 2.12 2026-03-20
2.14 TASK-R Implementing R... 2026-03-20 8d 2.13 2026-03-27
Add a task | Add a milestone
3 Testing 2026-03-27 22d 2026-04-17
3.1 TASK-S Integrate and tes... 2026-03-27 8d 2.14 2026-04-03
3.2 TASK-T Final testing and ... 2026-04-03 8d 3.1 2026-04-10
3.3 TASK-U Preparation for ... 2026-04-10 8d 3.2 2026-04-17
3.4 End of Testing and Imple... 2026-04-17 3.3 2026-04-17
Add a task | Add a milestone
Testing : 2026-03-27 - 2026-04-17

11. Milestones

Major Activity or Milestone	Estimated Milestone Target date	Owner/Reviewer Team Members
Deciding on tech stack finalized	November 3, 2025	Bryan Paul Yabut, Patrick Sherwin Millares
Data storage structure planned (local vs database)	November 4, 2025	Bryan Paul Yabut, Patrick Sherwin Millares
User roles and permissions completed	November 5, 2025	Patrick Solamillo, Ibrahim Abdillahi
Wireframes created for Admin site and Mobile app	November 13, 2025	Team
Database schema and relationships finalized	November 17, 2025	Bryan Paul Yabut, Patrick Solamillo, Yousuf Salihi
GitHub repositories setup and project initialized	January 12, 2026	Bryan Paul Yabut, Patrick Solamillo
Database schema rework and finalization	February 1, 2026	Bryan Paul Yabut, Patrick Solamillo
Database connection setup and seeded	February 3, 2026	Patrick Solamillo, Bryan Paul Yabut
Back-end logic and validations implemented	February 5, 2026	Patrick Solamillo, Bryan Paul Yabut, Patrick Sherwin Millares
Authentication System Implemented	February 7, 2026	Patrick Solamillo
Backend CRUD operations implemented	February 9, 2026	Patrick Solamillo, Bryan Paul Yabut, Patrick

		Sherwin Millares
Mobile app UI simplified and designed	February 12, 2026	Yousuf Salihi, Ibrahim Abdillahi
Scheduling, meal planning, and maintenance request features implemented	March 1, 2026	Patrick Sherwin Millares, Bryan Paul Yabut, Patrick Solamillo
Chore Manager and CSV export completed	March 5, 2026	Bryan Paul Yabut, Patrick Solamillo
Creation of API documentation for Mobile app team	March 10, 2026	Patrick Solamillo, Patrick Sherwin Millares, Bryan Paul Yabut
Budget Management data visualization added	March 15, 2026	Yousuf Salihi, Ibrahim Abdillahi
Mobile app UI completed	March 20, 2026	Yousuf Salihi, Ibrahim Abdilahi
REST API integrated with mobile app	March 27, 2026	Team
Unified dashboard integration	April 3 2026	Team
Final testing and debugging	April 10, 2026	Team
Final presentation and demo	April 17, 2026	Team

12. RAM – Responsibility Assignment Matrix

Create a RAM from your Task Listing. A sample is shown below:

Project Team Responsibilities					
Project Name:			**Dormio**		
Project Manager:			**Laily Ajellu**		
Task:	**Bryan Paul Yabut**	**Ibrahim Abdill**	**Patrick Sherwin Mill**	**Patrick Solamil**	**Yousuf Salih**
Deciding on tech stack finalized	P		S		
Data storage structure planned (local vs database)	P		S		
User roles and permissions completed		P		S	
Wireframes created for Admin site and Mobile app	S	S	S	S	P
Database schema and relationships finalized	P			S	S
GitHub repositories setup and project	P			S	
Database schema rework and finalization	S			P	
Database connection setup and seeded	S			P	
Back-end logic and validations implemented	S		P	S	
Authentication System Implemented				P	
Backend CRUD operations implemented	P		S	S	
Mobile app UI simplified and designed		S			P
Scheduling, meal planning, and maintenance request features implemented	P		S	S	
Chore Manager and CSV export completed	P		S	S	
Creation of API documentation for Mobile app team	S		P	S	
Budget Management data visualization added		S			P
Mobile app UI completed		S			P
REST API integrated with mobile app	P	S	S	S	S
Unified dashboard integration	S	S	S	P	S
Final testing and debugging	S	S	S	P	S
Final presentation and demo	S	S	S	S	P
P = Primary (Responsibility)	**S = Secondary (Support)**				

13. Approval

The signatures below indicate their approval of the contents of this document.

Project Role	Name	Signature	Date
Front-end Developer	Yousuf Salihi		2/5/2026
Back-end Developer	Patrick Solamillo		2/5/2026
Back-end Developer	Bryan Paul Yabut		2/5/2026
Back-end Developer	Patrick Millares		2/5/2026
Front-end Developer	Ibrahim Abdillahi		2/5/2026